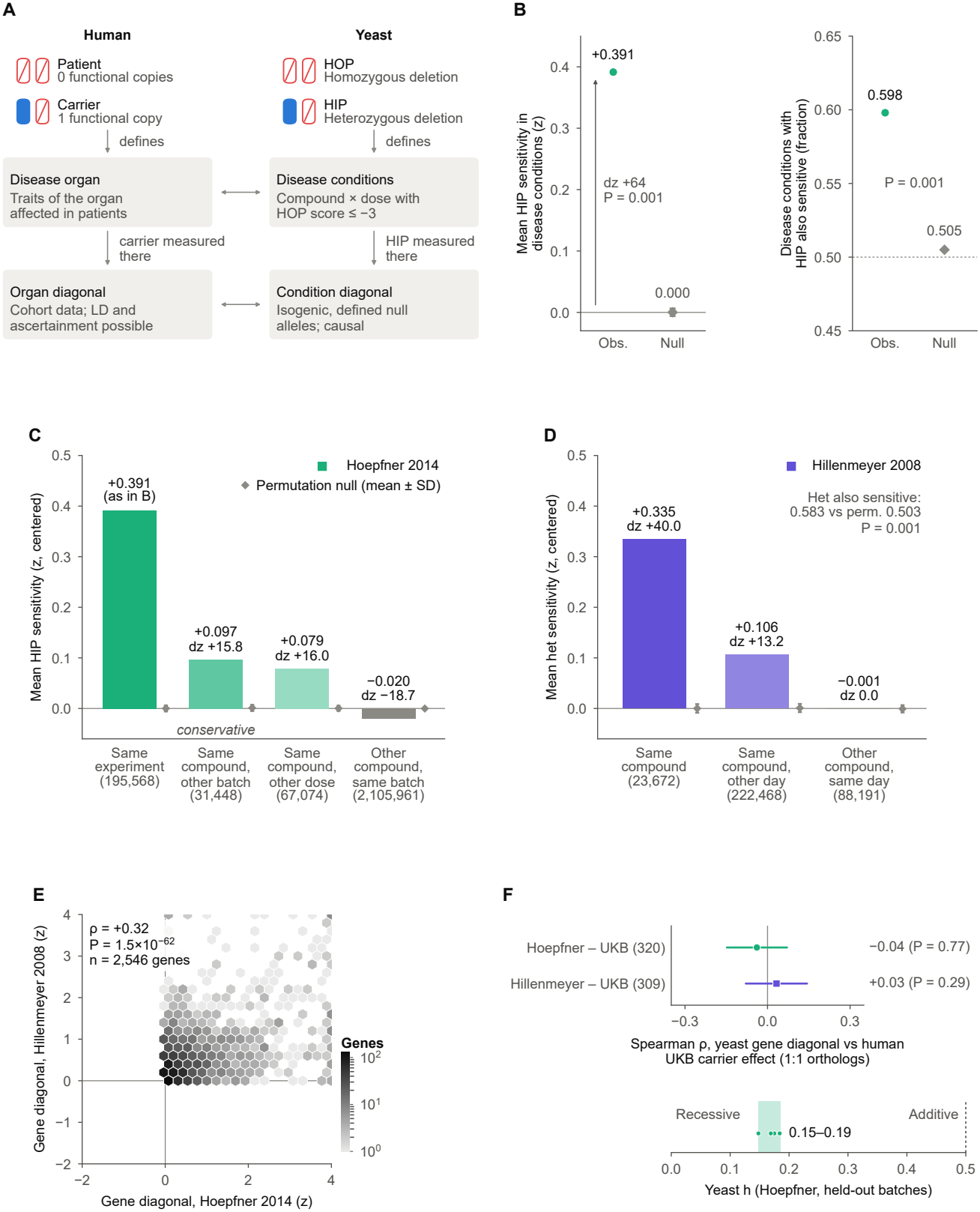

A
Human
Yeast
Patient
0 functional copies
Carrier
1 functional copy
defines
HOP
Homozygous deletion
HIP
Heterozygous deletion
defines
Disease organ
Traits of the organ affected in patients
Disease conditions
Compound × dose with HOP score ≤ −3
carrier measured there
HIP measured there
Organ diagonal
Cohort data; LD and ascertainment possible
Condition diagonal
Isogenic, defined null alleles; causal
B
Mean HIP sensitivity in disease conditions (z)
+0.391
dz +64
P = 0.001
0.000
Obs.
Null
Disease conditions with HIP also sensitive (fraction)
0.598
P = 0.001
0.505
Obs.
Null
C
Mean HIP sensitivity (z, centered)
Hoepfner 2014
Permutation null (mean ± SD)
+0.391
(as in B)
+0.097
dz +15.8
+0.079
dz +16.0
−0.020
dz −18.7
conservative
Same experiment (195,568)
Same compound, other batch (31,448)
Same compound, other dose (67,074)
Other compound, same batch (2,105,961)
D
Mean het sensitivity (z, centered)
Hillenmeyer 2008
Het also sensitive:
0.583 vs perm. 0.503
P = 0.001
+0.335
dz +40.0
+0.106
dz +13.2
−0.001
dz 0.0
Same compound (23,672)
Same compound, other day (222,468)
Other compound, same day (88,191)
E
ρ = +0.32
P = 1.5×10−62
n = 2,546 genes
Genes
Gene diagonal, Hillenmeyer 2008 (z)
Gene diagonal, Hoepfner 2014 (z)
F
Hoepfner – UKB (320)
−0.04 (P = 0.77)
Hillenmeyer – UKB (309)
+0.03 (P = 0.29)
Spearman ρ, yeast gene diagonal vs human UKB carrier effect (1:1 orthologs)
Recessive
Additive
0.15–0.19
Yeast h (Hoepfner, held-out batches)